

WORK DONE

Person B - Hybrid retrieval and MTEB submission harness | 25 September 2026

STATUS	BRANCH	REPOSITORY
Implemented, tested, committed and pushed	Atharva	github.com/AyushSinha2210/Team_Aftermath

Assigned scope and delivery

The two supplied project plans assign Person B the hybrid retriever and submission harness. The user requested implementation, separate commits, and a push to the existing Atharva branch. The delivered code covers all six planned retrieval modules, with an additional MTEB hybrid-search adapter and contract tests.

DELIVERABLE	IMPLEMENTED RESULT
Dense retrieval	Cosine search over Person A's encode(texts) API, with automatic pretrained fallback when its checkpoint is absent.
Sparse retrieval	Code-aware BM25 tokenization and an inverted index for CPU search.
Fusion and live pipeline	Weighted reciprocal rank fusion; HybridPipeline.retrieve(query, k=50) returns ranked snippets, scores and source attribution for Person C.
MTEB integration	PrePostPipelineEncoder(AbsEncoder) for dense embeddings; SearchProtocol adapter for actual dense + BM25 + RRF ranking.
Evaluation runner	Official AppsRetrieval test runner. It writes JSON only after MTEB returns a completed test result.

Verification

CHECK	EVIDENCE
Retrieval tests	7 passed on 25 September 2026, including the MTEB corpus text-field regression test.
Artifact integrity	Committed JSON identifies AppsRetrieval, test split, MTEB 2.21.3 and dataset revision f22508f96b7a36c2415181ed8bb76f76e04ae2d5.
Git remote	At audit, local HEAD and origin/Atharva both resolved to 2b93feecb1b025f219af7bab49cb2c4f10e14398; working tree clean.

MEASURED BASELINE

Official MTEB AppsRetrieval test result

0.06452	0.05089	0.10890
NDCG@10	MRR@10	Recall@10

Run context

ITEM	VALUE
Model	Pretrained sentence-transformers/all-MiniLM-L6-v2 fallback; no fine-tuned checkpoint was present.
Ranking	Dense cosine + code-aware BM25 + reciprocal rank fusion. Cross-encoder reranking was not used.
Task	MTEB AppsRetrieval test split; 3,765 queries and 8,765 corpus snippets.
Recorded evaluation time	1,587.782 seconds (approximately 26.5 minutes), as recorded in the result JSON.
Result	ariadne/submission/appsretrieval_results.json; context in ariadne/submission/release_notes.md.

Commit trail

a308f1c - hybrid retrieval pipeline; 14f06f7 - MTEB harness; 817e435 - batching and faster search; ec2ce0c - correct AppsRetrieval corpus field; 2b93fee - measured baseline and release notes.

Handoff and limits

Person B's assigned code and baseline are complete against the inputs available in this checkout. The published score is a baseline, not a fine-tuned or reranked final submission. Person A still needs to supply a checkpoint; Person C owns the valid-split configuration comparison and final selection. After that decision, the official test JSON and release notes should be regenerated together.

The first zero-score attempt exposed a corpus field mismatch: this task supplies code in text, not body. That invalid output was removed, the adapter and test were corrected, and the reported score is from the completed corrected run.

Scope references: ariadne-4person-execution-plan.pdf and prism-theme1-masterplan.pdf (supplied with the task).